Dr. Md Nadim

Curriculum Vitae

Faculty Member
Dept. of CSE, HSTU, Dinajpur-5200, Bangladesh
(+880) 1568260464, (+880) 1792550756
mnadims@hstu.ac.bd, mnadims.cse@gmail.com
Github *Linkedin*

Career Objective

To secure a research-intensive postdoctoral position where I can extend my work in quantum and classical machine learning and software engineering, with a focus on AI-driven practical applications. I aim to develop reliable and high-quality intelligent systems, collaborate with academic and industry partners, and translate advanced learning methods into real-world solutions that improve the lives of people who need them most.

Research Interests

- **Empirical Software Engineering & Reliable AI Systems:** Study software quality, security, and scalability in both human-written and AI-generated code, with a focus on building trustworthy systems for real-world domains such as healthcare and safety-critical applications.
- **AI for Health & Predictive Analytics:** Apply machine learning and deep learning methods to health and mHealth data to support early prediction, personalized care, and better decision making in clinical settings.
- **Quantum-Enhanced Software Analytics:** Design hybrid quantum and classical machine learning models to handle complex software engineering tasks, with the goal of improving performance and enabling new solutions for large-scale and data-intensive problems.

Education

2020–2025: **PhD, Computer Science**, *Software Research Lab*, University of Saskatchewan, Canada.
Software bug prediction using Quantum and Classical machine learning algorithms, their execution, comparison, and challenges. Utilised IBM Quantum services, D-Wave Quantum Annealing systems, and large language models for experimenting with QML algorithms in real quantum machines and simulators.

2018–2020: **MSc, Computer Science**, *Software Research Lab*, University of Saskatchewan, Canada.

2006–2010 : **BSc, Computer Science & Engineering**, *Hajee Mohammad Danesh Science & Technology University*, Dinajpur, Bangladesh.

Publications: Journal Articles

2025 **Md Nadim**, Mohammad Hassan, Ashis Kumar Mandal, Chanchal K. Roy, Banani Roy, and Kevin A. Schneider. Comparative analysis of **quantum and classical support vector classifiers** for software bug prediction: an exploratory study. *Quantum Machine Intelligence*, volume 7, page 32, Mar 2025.

2025 Ashis Kumar Mandal, **Md Nadim**, Chanchal K. Roy, Banani Roy, and Kevin A. Schneider. **Quantum software engineering and potential of quantum computing** in software engineering research: a review. *Automated Software Engineering*, volume 32, page 27, Mar 2025.

2023 **Md Nadim** and Banani Roy. Utilizing source code syntax patterns to detect bug inducing commits using machine learning models. *Software Quality Journal*, volume 31, pages 775–807, Sep 2023.

2022 **Md Nadim**, Manishankar Mondal, Chanchal K. Roy, and Kevin A. Schneider. Evaluating the performance of clone detection tools in detecting cloned co-change candidates. *Journal of Systems and Software*, volume 187, page 111229, 2022.

2022 **Md Nadim**, Debajyoti Mondal, and Chanchal K. Roy. Leveraging structural properties of source code graphs for just-in-time bug prediction. *Automated Software Engineering*, volume 29, page 27, Mar 2022.

Publications: Conference Proceedings

2025 **Md Nadim**, Mohammad Hassan, Ashis Kumar Mandal, and Chanchal K. Roy. **Quantum Vs. Classical Machine Learning Algorithms** for Software Defect Prediction: Challenges and Opportunities . In *IEEE/ACM International Workshop on Quantum Software Engineering (Q-SE)*, pages 35–42, Ottawa, Canada, May 03, 2025. IEEE Computer Society.

2024 Ashis Kumar Mandal, **Md Nadim**, Chanchal K. Roy, Banani Roy, and Kevin A. Schneider. Evaluating the performance of a **D-Wave Quantum Annealing System** for feature subset selection in software defect prediction. In *IEEE International Conference on Quantum Computing and Engineering (QCE)*, volume 02, pages 103–108, 2024.

2020 **Md Nadim**, Manishankar Mondal, and Chanchal K. Roy. Evaluating performance of clone detection tools in detecting cloned cochange candidates. In *IEEE 14th International Workshop on Software Clones (IWSC)*, pages 15–21, 2020.

2016 Shahnaj Parvin Shathi, Md. Delowar Hossain, **Md Nadim**, Sayed Golam Rasul Riayadh, and Tangina Sultana. Enhancing performance of naïve bayes in text classification by introducing an extra weight using less number of training examples. In *2016 International Workshop on Computational Intelligence (IWCI)*, pages 142–147, 2016.

2010 Ashis Kumar Mandal, Md. Delowar Hossain, and **Md Nadim**. Developing an efficient search suggestion generator, ignoring spelling error for high speed data retrieval using double metaphone algorithm. In *13th International Conference on Computer and Information Technology (ICCIT)*, pages 317–320, 2010.

Research Experience: University of Saskatchewan, Canada

Sep, 2018 – Apr, 2025 ***Graduate Researcher***.
I conduct research on software defect prediction and code clone detection, with a focus on using machine learning, deep learning, and quantum-inspired methods. My work explores source code structure, syntax patterns, and feature learning to improve bug prediction and software quality.

Advisor : **Dr. Chanchal K. Roy**, *Professor, Department of Computer Science*, University of Saskatchewan, Canada (*Personal Web-page*)

Teaching Experience

May 2025 – Present **Associate Professor, Department of CSE**, *Hajee Mohammad Danesh Science and Technology University (HSTU)*, Dinajpur, Bangladesh.
Courses Taught: Software Engineering (CSE-305); Fundamentals of Computer & Programming (CSE-101/102/107/108/132); Mathematical Analysis for Computer Science (CSE-361); Advanced Machine Learning (CSE-547)

2018 – 2025 **Study Leave, M.Sc. and Ph.D.**, *University of Saskatchewan*, Saskatoon, Canada.
September 1, 2018 to May 14, 2025

Feb 2012 – Aug 2018 **Lecturer/ Assistant Professor, Department of CSE**, *HSTU*, Dinajpur, Bangladesh.
I joined as a Lecturer on February 1, 2012, and was promoted to Assistant Professor on February 1, 2015. During this time, I continued teaching key undergraduate courses at HSTU, including Structured Programming (C), Object-Oriented Programming (C++ and Java), Data Structures, Algorithms, Operating Systems, Database Management Systems, and Pattern Recognition.

Teaching Assistantship @USASK, SK, Canada

Winter, 2024 **CMPT 340: Programming Language Paradigms**, *132 Hours*.

Fall, 2023 **CMPT 214: Programming Principles and Practice**, *88 Hours*.

Spring, 2023 **CMPT 145: Principles of Computer Science**, *66 Hours*.

Winter, 2023 **CMPT 280: Intermediate Data Structures and Algorithms**, *88 Hours*.

Winter, 2022 **CMPT 141: Introduction to Computer Science**, *88 Hours*.

Winter, 2021 **CMPT 470: Advanced Software Engineering**, *88 Hours*.

Fall, 2020 **CMPT 214: Programming Principles and Practice**, *88 Hours*.

Research Collaborations

I worked jointly with researchers from different well-known universities and research organizations on a number of research projects. In these collaborations, I took the main responsibility for developing the research concepts, planning and running the experiments, analyzing findings, and preparing the research articles. The collaborative discussions and expert feedback from my co-researchers helped improve the quality, clarity, and impact of the work. My collaborators include:

2023 – 2025 **Dr. Mohammad Mahdi Hassan**, *Associate Professor*, Faculty of Sciences and Engineering, East West University, Bangladesh.

I collaborated with Dr. Hassan on research exploring Quantum Machine Learning (QML) for software defect prediction and buggy commit detection. I contributed to the research design, implementation of QML and classical ML models, experimental analysis, and manuscript preparation. This collaboration resulted in two publications, one in the **Quantum Machine Intelligence Journal** in March 2025, and another in the **IEEE/ACM International Workshop on Quantum Software Engineering (Q-SE)** in May 2025.

2021 – 2022 **Dr. Banani Roy**, *Associate Professor*, University of Saskatchewan, Canada.

I collaborated with her on research investigating source code syntax patterns for Machine Learning based buggy commit detection. I contributed to the feature engineering approach, experimental design, dataset analysis, evaluation of multiple ML models, and manuscript preparation. The study demonstrated that syntax pattern based features can improve both the performance and explainability of buggy commit prediction models. This work was published in the **Software Quality Journal** in December 2022.

2021 – 2022 **Dr. Debajyoti Mondal**, *Associate Professor*, University of Saskatchewan, Canada.

I collaborated with him on research related to graph-based source code analysis for Just-in-Time (JIT) buggy commit prediction. I contributed to the research design, development of the Source Code Graph (SCG) methodology, feature extraction, large-scale experimental analysis, and manuscript preparation. This work was published in the **Automated Software Engineering Journal** in March 2022.

2018 – 2022 **Dr. Manishankar Mondal**, *Professor*, Khulna University, Bangladesh.

I collaborated on a research project focused on detecting cloned co-change candidates in evolving software systems. My contribution included analyzing clone detection techniques, conducting experiments with multiple clone detection tools on open-source software systems, processing and evaluating experimental results, and assisting in the preparation of the research manuscript. The collaboration also involved technical discussions on software evolution, code clone analysis, and change impact analysis to improve the study's overall quality and direction. The research outcomes from this collaboration were published in the IEEE **International Workshop on Software Clones** (IWSC'2020). An extended version of the work was later published in the **Journal of Systems and Software** (JSS) in 2022.

2010 – **Dr. Ashis Kumar Mandal**, *Professor*, CSE, HSTU, Dinajpur, Bangladesh.

I collaborated with Dr. Mandal on several research projects spanning information retrieval, software defect prediction, feature selection, Quantum Machine Learning (QML), and Quantum Software Engineering (QSE). In these collaborations, I contributed to the development of research ideas, experimental design, implementation of machine learning and quantum computing techniques, dataset analysis, performance evaluation, and manuscript preparation. Our collaborative work resulted in publications in leading venues, including the IEEE International Conference on Quantum Computing and Engineering (QCE), IEEE Access Journal, Quantum Machine Intelligence Journal, Automated Software Engineering Journal, IEEE/ACM International Workshop on Quantum Software Engineering (Q-SE), and the International Conference on Computer and Information Technology (ICCIT) between 2010 and 2025.

Fellowships & Awards

2018 Dean's Scholarship, University of Saskatchewan

2010 Prime Minister Gold Medal, Bangladesh

Leadership and Community Service

Jan 21, 2024 Crowd Support (Watch Party), TEDx USASK, Canada

2020–2021 Vice President, CSGC, USASK, Canada

2019–2020 GSA Representative, CSGC, USASK, Canada

May 2019 Volunteer Lead (Data Science Workshop), DIGITIZED, USASK, Canada

Administrative Roles

2025 **Member, Executive Committee**, Faculty of Computer Science & Engineering, HSTU, Dinajpur-5200, Bangladesh

2015 **Chairman**, Department of Computer Science & Information Technology (CIT), HSTU, Dinajpur-5200, Bangladesh

Key Technical Skills

Programming: Python, C, C++, Java

Tools: Machine Learning, Deep Learning, Quantum Computing, High-Performance Computing, NumPy, Pandas, Scikit-learn, TensorFlow/PyTorch, Git

Quantum Algorithms: Phase Estimation, Amplitude Amplification, Grover's Search, Shor's Algorithm, Hamiltonian Simulation, Qubitization, Quantum Error Correction (QEC)

Web & DB: HTML5, JavaScript, PHP, SQL, MySQL

Scripting: Linux, Shell Scripting

Certification Courses

2024 **Graduate Professional Skills Certificate @University of Saskatchewan**, *SK*, Canada.
The Graduate Professional Skills Certificate is a comprehensive, non-credit program for graduate students and postdoctoral researchers at the University of Saskatchewan, SK, Canada. Within this Certificate program, GPS 974 focuses on strength-based professional skills development, building reflective practice, and developing a professional portfolio for improving skills and growth.

2024 **Thinking Critically@University of Saskatchewan**, *SK*, Canada.
The goal of this class is to provide a supportive and challenging setting to develop the creative and critical thinking skills required for professional practice. GPS 984 focuses on foundational frameworks of thinking (often invisible to us) that are used for almost everything we do in our personal and professional lives.

Presentations/ Invited Talks/ Seminars

Jan 19, 2026 **Trainer: AI & Cyber Security Fundamentals**, *Rangpur Regional Office of Bangladesh Computer Council*, Bangladesh.
I delivered foundational training on the safe use of Artificial Intelligence (AI) and Cyber Security to government officials from various sectors of the Bangladesh Civil Service.

May 03, 2025 **Conference Presentation @Q-SE, Co-located with ICSE'2025**, *Ottawa*, Canada.
I delivered a research talk on the comparative analysis of Quantum Machine Learning (QML) and Classical Machine Learning (CML) algorithms for software defect prediction.

Jan 15, 2025 **Ph.D. Research Seminar @University of Saskatchewan**, Canada.
I delivered a research seminar at the University of Saskatchewan (USASK), Canada, where I presented the major research contributions and findings from my Ph.D. work to faculty members and graduate students.

Sep 16, 2024 **Presenting Research Paper @IEEE Quantum Week**, *Montréal Convention Centre*, Canada.
I delivered a research talk on the application of quantum annealing for feature subset selection in software defect prediction. The presentation highlighted the use of the D-Wave Quantum Processing Unit (QPU) to optimize feature selection and compared its effectiveness with classical approaches using multiple software defect datasets.

Sep 12, 2024 **Quantum Software Engineering Talk @University of Saskatchewan**, Canada.
I delivered a technical talk on quantum computing and quantum annealing for software defect prediction, where I introduced the fundamentals of quantum computing, D-Wave quantum annealing systems, and practical optimization techniques. The presentation also demonstrated our research on QA-based feature subset selection for software defect prediction, including QUBO formulation, Python-based implementation, experimental evaluation on AEEM, JIRA, and NASA datasets, and comparative analysis with classical approaches.

Jun 12-13, 2024 **Poster Presentation @Software Engineering for Machine Learning Applications (SEMLA)**, *Polytechnique Montréal*, Canada.
Presented an experimental comparison between quantum and classical machine learning approaches for software defect prediction

Feb 18, 2020 **Research Paper Presentation @IEEE 14th International Workshop on Software Clones (IWSC)**, *London, ON*, Canada.
I delivered a research talk on evaluating clone detection tools for identifying cloned co-change candidates during software evolution.

Jun 12, 2014 **Invited Talk on Freelancing @District Commissioner's Office**, *Dinajpur*, Bangladesh.

Referees

Dr. Chanchal K. Roy
Professor
Department of Computer Science
University of Saskatchewan, Canada
☎ +1 (306) 966-4163
✉ chanchal.roy@usask.ca
https://clones.usask.ca

Dr. Md. Abdulla Al Mamun
Professor & Chairman, Department of Computer Science & Engineering (CSE)
HSTU, Dinajpur-5200, Bangladesh
✉ mamun@hstu.ac.bd